

Do We Need To Worry About Radiation Emitting Devices In Common Use?

Digitalisation of resources has made the use of devices like phones and laptops extremely frequent. However, the excess use of such devices can cause harmful effects since these also emit electromagnetic radiations. So, it's important for consumers to take necessary steps to ensure that they are not exposed to radiations round the clock

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From kids to adults, be it for education or work, digitalisation of resources has made the use of devices like phones and laptops extremely frequent. Even for recreational activities, we have moved to such devices for experiencing everything from the comfort of our homes. In today's world, it is a fact that we are surrounded by technology everywhere to such an extent that we don't even realise it. But with the excess use comes rising concerns.

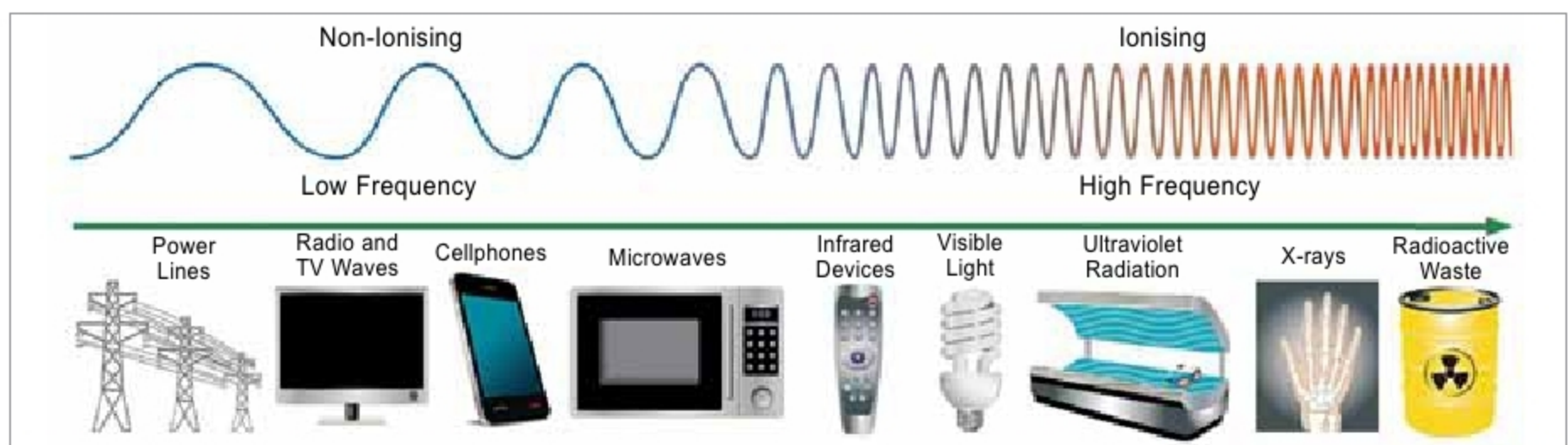
We would often hear parents telling their kids things like 'don't hold the phone too close' or 'don't keep the laptop on your lap.' The reason is simple—improper and excess use of even the best technologies causes acute and chronic harmful effects, with problems like dry eyes and headaches being the most common. The fear is only aggravated by reports that state electromagnetic radiations like ionising ultraviolet or X-rays can cause DNA damage and have long term risks associated with serious

problems like cancer since the gadgets in our daily use also emit electromagnetic radiations.

About RF electromagnetic radiation

In the context of commonly used electronic devices, electromagnetic radiation is in the form of radio frequency (RF) energy waves. When alternating electric current travels through the antenna, electromagnetic waves are emitted and received by the antennas inside these devices. Higher the frequency, the faster is the data transfer. Metal parts in the machines amplify RF signals.

Since wireless devices have to detect other tools that they wish to connect to, they increase their emitting power to find them. The permissible upper limit for wireless devices set by the US Federal Communications Commission (FCC) is 1.6 W/kg of RF radiation. Some devices that are commonly used and emit electromagnetic radiations are discussed next.



RF radiation in different devices in order of increasing frequency in the electromagnetic spectrum (Credit: www.niehs.nih.gov)